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REPORT DOCUMENTATION PAGE

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1. REPORT NUMBER: TOP 6-2-185	2. GOVT ACCESSION NO.: A141706
3. RECIPIENT'S CATALOG NUMBER:	
4. TITLE *(and Subtitle)*: US ARMY TEST AND EVALUATION COMMAND TEST OPERATIONS PROCEDURE METEOROLOGICAL SOUNDING SYSTEMS	5. TYPE OF REPORT & PERIOD COVERED: Final
	6. PERFORMING ORG. REPORT NUMBER:
7. AUTHOR(s):	8. CONTRACT OR GRANT NUMBER(s):
9. PERFORMING ORGANIZATION NAME AND ADDRESS: US ARMY ELECTRONIC PROVING GROUND (STEEP-MT-T) FORT HUACHUCA, ARIZONA 85613	10. PROGRAM ELEMENT, PROJECT, TASK AREA & WORK UNIT NUMBERS: DARCOM-R 310-6
11. CONTROLLING OFFICE NAME AND ADDRESS: US ARMY TEST AND EVALUATION COMMAND (DRSTE-AD-M) ABERDEEN PROVING GROUND, MARYLAND 21005	12. REPORT DATE: 15 May 1984
	13. NUMBER OF PAGES: 38
14. MONITORING AGENCY NAME & ADDRESS *(if different from Controlling Office)*:	15. SECURITY CLASS. *(of this report)*: Unclassified
	15a. DECLASSIFICATION/DOWNGRADING SCHEDULE:

16. DISTRIBUTION STATEMENT *(of this Report)*

Approved for public release; distribution unlimited.

17. DISTRIBUTION STATEMENT *(of the abstract entered in Block 20, if different from Report)*

18. SUPPLEMENTARY NOTES

19. KEY WORDS *(Continue on reverse side if necessary and identify by block number)*

Meteorological Sounding Systems
NFIM
Direction Finder
Radiosonde
Balloon
Humidity Sensor

20. ABSTRACT *(Continue on reverse side if necessary and identify by block number)*

This test operations procedure (TOP) delineates general test and specific subtest procedures required to measure the technical performance of Meteorological Sounding Systems. Comparison of test results with criteria and technical specifications permits evaluation of their suitability for an intended use. The test procedures contained herein apply only to meteorological sounding systems which determine atmospheric pressure, temperature, humidity and upper atmospheric wind speed and direction. Testing of rocket carrier vehicles is not included.

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US ARMY TEST AND EVALUATION COMMAND

TEST OPERATIONS PROCEDURE

DTSTE-RP-702-105 15 May 1984
Test Operations Procedure 6-2-185*
AD No. A141706

METEOROLOGICAL SOUNDING SYSTEMS

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1. SCOPE. This test operations procedure (TOP) delineates general test and specific subtest procedures for measuring and evaluating the technical performance and characteristics of Meteorological Sounding Systems relative to criteria specified in the Required Operational Capability (ROC), Letter Requirements (LR), and Coordinated Test Program (CTP). The test procedures contained herein apply only to meteorological sounding systems (GMD) which determine atmospheric pressure, temperature, humidity, and upper atmospheric wind speed and direction. Testing of rocket carrier vehicles and LORAN or OMEGA positioning-type systems are not included. This TOP is limited to tests outlined herein. The following tests will measure:

a. Radiosonde Power Output - The objective of this subtest is to determine the power output of the radiosonde transmitter.

b. Radiosonde Battery Life - The objective of this subtest is to determine the life of the batteries used in the radiosonde test item.

c. Radiosonde Transmitter Range Test - The objective of this subtest is to determine the radiosonde's transmitter range.

d. Sensor Response and Temperature-Humidity and Pressure-measuring accuracy - The objective of this subtest is to determine the sensor response and the accuracy with which the radiosonde measures temperature, pressure and humidity.

e. Upper Wind Speed and Direction Measuring Accuracy - The objective of this subtest is to determine the accuracy with which the system measures upper wind speed and direction.

*This TOP supersedes MTP 6-2-185, 15 December 1969.

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f. Frequency Accuracy and Stability - The objective of this subtest is to determine the frequency stability and accuracy of the radiosonde transmitter and ground equipment.

g. Receiver Sensitivity - The objective of this subtest is to determine the sensitivity of the receiver by measuring the power level of the Minimum Discernible Signal (DMS).

h. Receiver Selectivity - The objective of this subtest is to determine the frequency response curve of the receiver.

i. Recording Equipment Test - The objective of this subtest is to determine the characteristics of the systems recorder (if applicable).

j. Computer Test - The objective of this subtest is to determine the characteristics and performance of the test item's computer system.

k. Antenna Test - The objective of this subtest is to determine the performance of the test item's antenna.

l. Balloon Test - The objective of this subtest is to determine the performance of the balloon.

m. Systems Test - The objective of this subtest is to determine the overall performance of the test item.

2. FACILITIES AND INSTRUMENTATION.

2.1 Facilities. Tests shall be performed at a laboratory with environmental chambers providing controlled temperature, humidity, and pressure (altitude).

2.2 Instrumentation and Equipment. The following instrumentation and equipment will be required to support this TOP.

a. Instrumented temperature, humidity, and pressure chambers.

b. Power meter.

c. Frequency counter.

d. Oscilloscope.

e. Voltmeter.

f. Spectrum analyzer.

g. Signal generator.

h. Theodolite or surveyor's transit.

i. Dummy load equipment.

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j. Signal sampler.

k. Attenuator.

l. Noise and field intensity meter (NFIM).

m. Distortion analyzer.

2.3 Characteristics/Requirements. The characteristics and accuracies required of the above test instrumentation are determined by the performance specifications of the individual meteorological sounding system to be tested. Select test instrumentation having an accuracy of sufficiently higher quality (10:1) than that of the test item. Calibration of all test instrumentation shall be traceable to the National Bureau of Standards. The above listed major facilities, instrumentation and equipment will provide the necessary characteristics and setups required to perform the subtests outlined by this TOP.

3. PREPARATION FOR TEST

a. Review TECOM Pamphlet 70-3, Research, Development, and Acquisition and Project Engineer's Handbook for guidance on test planning, execution, reporting and post-test activities.

b. Establish and/or continually maintain a readily accessible project log and project file.

c. Review local installation project office handbook, standing operating procedures (SOP) and implementing directives which govern the administrative processes of preparing test plans, conducting tests, preparing reports, reporting to the Test Resources Management System (TRMS) and budgeting.

d. Acquire and review all descriptive, instructional, and specification material on the test items issued by Government and contractor(s) for checking the test plan's subtest objectives, criteria, facility(ies) and instrumentation requirements.

e. Determine the scheduled availability of the test item.

f. Ensure availability of appropriate facilities, and coordinate the test support requirements including personnel, equipment, maintenance, spare parts and instrumentation.

g. Review the detailed test plan.

h. Inventory the test items and test support items for the test.

i. Record as a minimum the following data:

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(1) Nomenclature, serial number(s), manufacturer's name, and function of the item(s) under test.

(2) Nomenclature, serial number, accuracy tolerance, calibration requirements, and last calibration data of test equipment selected for the tests.

(3) Damages to the test item(s) incurred during transit and/or manufacturing defects.

(4) Test item photographs.

j. Establish instrumentation or measurement system mean error and standard deviation of error.

k. Determine test item sample size.

4. TEST CONTROLS

a. Organize test team and establish responsibilities for test conduct, reporting, and data control.

b. Select test equipment having an accuracy at least 10 times greater than that of the function to be measured.

c. Familiarize all test personnel with the test item instructional material.

d. Prepare adequate safety precautions to provide safety for personnel and equipment and ensure all safety SOP's are observed throughout the test.

e. Inspect the test item thoroughly for obvious physical and electrical defects. Note and correct all defects before proceeding with testing.

f. Prepare, disseminate and continually review record forms for systematic entry of data, chronology of tests, and analysis in final evaluation of the test item.

g. Prior to beginning any subtest, verify that the equipment is aligned as specified in the pertinent operating instructions and represents average equipment in normal operating condition.

h. Ensure that appropriate security measures are instituted to safeguard classified material and data and that arrangements for supporting and participating agencies, activities, and facilities have been made.

i. Prepare and monitor a test item sample plan sufficient to ensure that enough samples of all measurements are taken to provide statistical confidence of final data in accordance with TOP 3-1-002, Confidence Intervals and Sample Size, dated 25 Jan 67, AD No. 718229.[5]

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5. PERFORMANCE TESTS

NOTE: Modification of these procedures shall be made as required by technical design of the test item and availability of test equipment, but only to the extent that such modified procedures will not affect the validity of test results.

5.1 Test Methods.

5.1.1 Radiosonde Power Output Test.

NOTE: The power output of the radiosonde transmitter is a good indication of its overall condition.

a. Connect the Radiosonde test item and the test instrumentation as shown in figure 1.

b. Turn on the Radiosonde and instrumentation and allow sufficient time for all items to warm up and stabilize. (Refer to the appropriate technical manuals.)

c. Tune the Radiosonde transmitter to a standard test frequency, in accordance with the tuning procedure outlined in the appropriate technical manual. The degree of coupling afforded by the signal sampler shall be such that the input to the Noise and Field Intensity Meter (NFIM) is limited to a millivolt or less.

NOTES:
1. This is not a test of the absolute power value as coupling losses are considered.
2. Since all emissions emanating from a transmitter are related to the fundamental output power, and since the fundamental output power can be expected to vary over the entire transmitter operating range, the power output tests shall cover the tuning range of the radiosonde transmitter.

d. Tune the NFIM to the Radiosonde's carrier frequency.

e. Note and record all indicator settings and attenuation factors used.

f. Substitute a signal from a signal generator tuned to the transmitter carrier frequency at the input to the NFIM. Adjust the output attenuator of the signal generator until the reference reading is again obtained on the NFIM.

g. Repeat the procedure of step e, above.

NOTE: The effects of decreasing battery voltage shall be determined in conjunction with the battery life test, paragraph 5.1.2.

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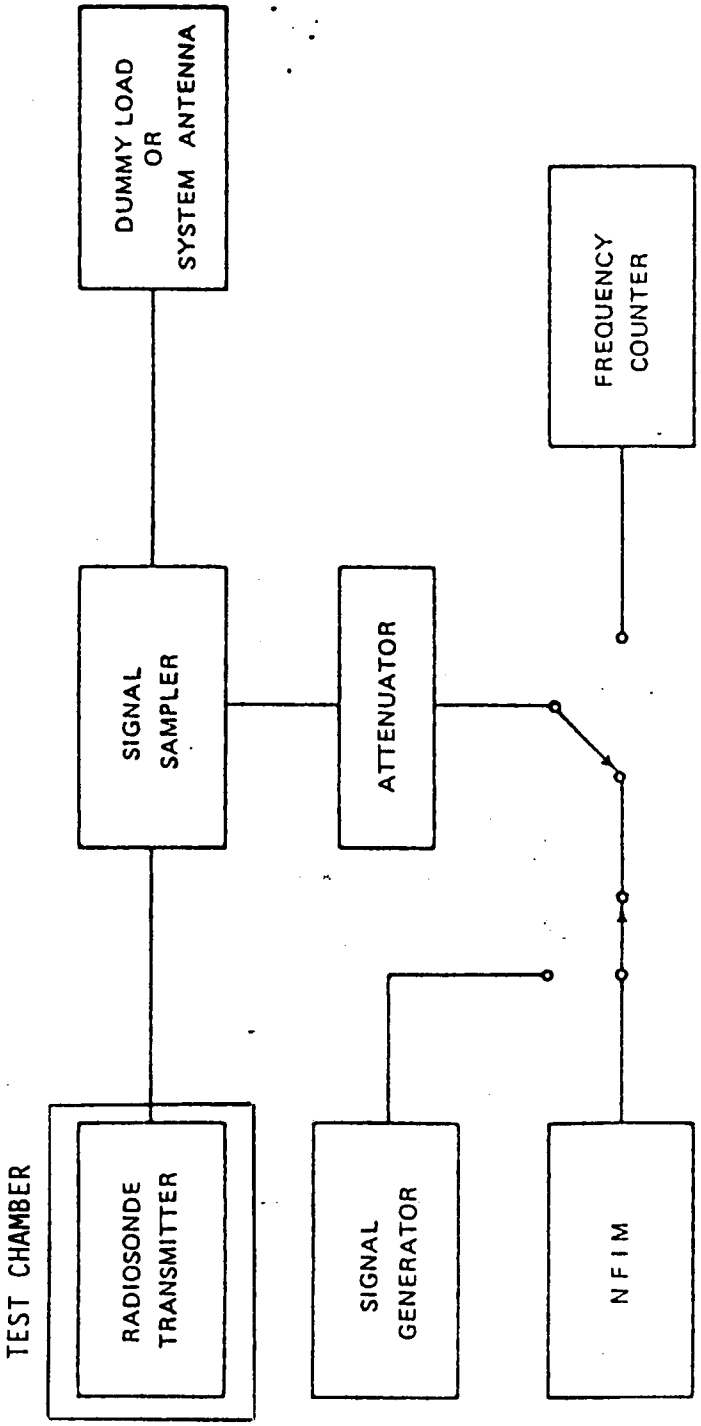


Figure 1. Radiosonde power output measurement block diagram.

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5.1.2 Radiosonde Battery Life Test.

NOTE: The radiosonde battery life shall be determined in conjuction with temperature-altitude environmental chamber sensor test.

a. Select three batteries at random and install them in a Radiosonde test item that has been packaged for normal flight.

b. Set the chamber temperature and altitude to simulate a 10-minute check at ambient conditions. Then simulate a one and a half hour flight to 100,000 feet.

c. Note and record battery voltage readings and power output at 5-minute intervals during step b, above, including the return cycle from 100,000 feet to ambient conditions.

d. Note and record the lowest voltage level for a usable signal.

e. Determine the power output from a radiosonde transmitter when the antenna is permanently secured to the transmitter.

5.1.3 Radiosonde Transmitter Range Test. Perform the radiosonde transmitter range test in accordance with manufacturers instructions, if available. If not, use TOP 6-2-242 Receiver-Transmitter, System Tests, 15 June 81, AD No. A100395.[6]

5.1.4 Sensor Response and Temperature, Humidity, and Pressure Accuracy Test.

5.1.4.1 General.

a. Conduct these tests under controlled conditions by comparison with calibrated laboratory standards (reference sensors).

b. Use pressure, temperature and humidity chambers that provide a range sufficient to cover the stated capability of the sounding element.

c. Use a fan to provide circulation of air for temperature and humidity tests.

d. Connect the Radiosonde output (by coaxial cable) to the receive-record unit.

NOTE: Since the transmitted value of the sensor "reading" depends upon the resistance versus frequency transfer characteristics of the Radiosonde, the uniformity of Radiosonde shall be checked during the test. Three or more Radiosondes of the same type shall be used on each chamber test. If sufficient receiving and recording equipment is not available, arrangements shall be made to record values alternately.

e. Adjust the Radiosonde to transmit one element (temperature or humidity) continuously during temperature and humidity tests.

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NOTES: 1. Since the recorder is used in this test the data may need adjustment depending upon the results of tests conducted in paragraphs 5.1.9 and 5.1.10.

2. Usually each Radiosonde is calibrated in a samll chamber prior to flight and a correction (if any) applied to all readings. However, for this test a preliminary correction will not be determined. The correction can be determined from the data.

5.1.4.2 Temperature Sensor.

a. Conduct this test under controlled temperature conditions (chamber) by comparison with a reference standard accurate to ±0.1 degree C.

b. Mount the reference sensor beside the Radiosonde temperature element. A low mass thermocouple with a rapid response is preferred.

c. Vary the chamber temperature in 5°C increments over the entire stated range of Radiosonde, from ambient temperature through the low temperature range, to the high range and return to ambient. Comparison readings shall be made each minute.

d. Install a small electrical heating element so heated air will flow over the sensors. Several times during the chamber test, apply current to the heating element for short periods and observe the response of the sensors. This will give an indication of the lag or response time of the temperature element.

e. If the Radiosonde temperature elements are replaceable, the test will be repeated with other elements.

5.1.4.3 Humidity Sensor.

a. Conduct this subtest under controlled-temperature and relative humidity conditions (chamber) by comparison with a reference standard accurate to ±1 percent.

b. Starting with dry air (approximate) and with the temperature held constant at the highest stated operating temperature range of the humidity element, increase the moisture content of the air in the chamber in increments of about 5 percent relative humidity to greater than 100 percent (supersaturation). Condensation should not occur on the element. Humidity changes should be made as rapidly as possible and comparative readings should be obtained as often as the standard sensor will permit. A timed recovery test from supersaturation back to 20 percent relative humidity should be included. This simulates conditions the sensor experiences going from cloud to clear air.

c. Repeat the readings at the low- and mid-temperature ranges of the humidity element.

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5.1.4.4 Pressure System.

a. Place the radiosonde in an altitude chamber (low pressure) with a calibrated aneroid barometer.

b. Compare the two readings and adjust the radiosonde element, using the calibration chart provided.

c. Decrease the pressure in the chamber in 100-millibar increments, indicated on the calibration chart, to the lowest operating range of the element.

d. Raise the pressure in the chamber in 100-millibar increments to the highest range of the pressure element and compare values with the standard unit.

e. Decrease the pressure in the chamber from the maximum pressure of the element in increments indicated on the calibration chart to the lowest operating range of the element.

f. Repeat procedure at least once to verify the durability of the sensor.

5.1.5 Upper Wind Speed and Direction Measurement Accuracy Test.

NOTE: One of the primary reasons for upper air soundings is to obtain wind data, which is determined from space positioning of the radiosonde or the sounding element. The space positioning is accomplished by radar and by Direction Finding (DF) radio gear. This test shall include minimum range and elevation angle for lock-on, low angle reflections and azimuth angle accuracy. The DF system characteristics shall be determined in accordance with TOP 6-2-070, Direction Finder Set, Radio dated 27 November 1968.

a. Determine the minimum range for lock-on by placing the sounding element on a movable cart 100 (30.5 m) feet from the receiver unit and moving towards it while transmitting. Observe the distance at which the receiver is locked on the Radiosonde signal.

b. Determine the minimum elevation angle for lock-on by raising and lowering the radiosonde on a pole while transmitting. The minimum of elevation angle for lock-on of the receiver and radiosonde shall be determined by means of a manually operated theodolite placed as near the ground unit as possible.

c. Determine the low angle reflection by raising then lowering a radiosonde on a captive balloon at less than 15° intervals horizontally around the ground unit and at two or three different ranges from the ground unit. Since the critical reflection angles usually are below 15° above the horizon, the tests should cover this angle as far from the ground unit as possible. The effect of low angles shall be noted and the exact angle measured with a manually operated theodolite.

NOTE: This test should be conducted during calm weather, no wind.

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d. Determine the azimuth angle accuracy by comparing the Radiosonde's azimuth position as indicated by the ground unit with the theodolite's azimuth angle indication, when performing step c, above.

5.1.6 Frequency Accuracy and Stability.

Perform the frequency accuracy and stability tests in accordance with TOP 6-2-517, Transmitter Frequency Accuracy and Stability, dated 31 December 1980, AD No. A100394 for the sounding element and ground equipment, as appropriate.

5.1.7 Receiver Sensitivity Test.

a. Connect the receiver under test and the test instrumentation as shown in figure 2. The test setup (fig. 2) shall be in a shielded room with the apparatus on a grounded metal bench top.

b. Terminate the receiver output in the specified load impedance, the value of which may be found in the technical manual for the equipment under test.

NOTE: The receiver sensitivity measurement yields an overall indication of the receiver's operating condition, alignment and tracking.

c. Tune the receiver and signal generator to a low frequency in the receiver's tuning range (low, medium and high test frequencies within the receiver's tuning range shall be selected in accordance with Measurement of Radio Frequency Spectrum Characteristics, MIL-STD-449D dated 22 February 1973).

NOTE: The type of modulation shall be the same as provided in the Radiosonde which is used with the system under test.

d. Set the signal generator modulation to approximately 30 percent for AM test items and make frequency deviation settings for FM test items.

e. Adjust the generator output so that the distortion analyzer shows a response at the sensitivity test output level as specified. Note the generator output level in dbm and the frequency.

f. Repeat the above procedure for each test frequency, including a specific measurement at the frequency of the sounding element used with the system.

5.1.8 Receiver Selectivity Test.

a. Connect the receiver and test instrumentation in the same manner as described in paragraph 5.1.7a (See fig. 2.), with the addition of a frequency meter, which is coupled to the signal generator by means of a resistive pad or attenuator. Care must be taken to ensure that the frequency meter does not also couple into the receiver.

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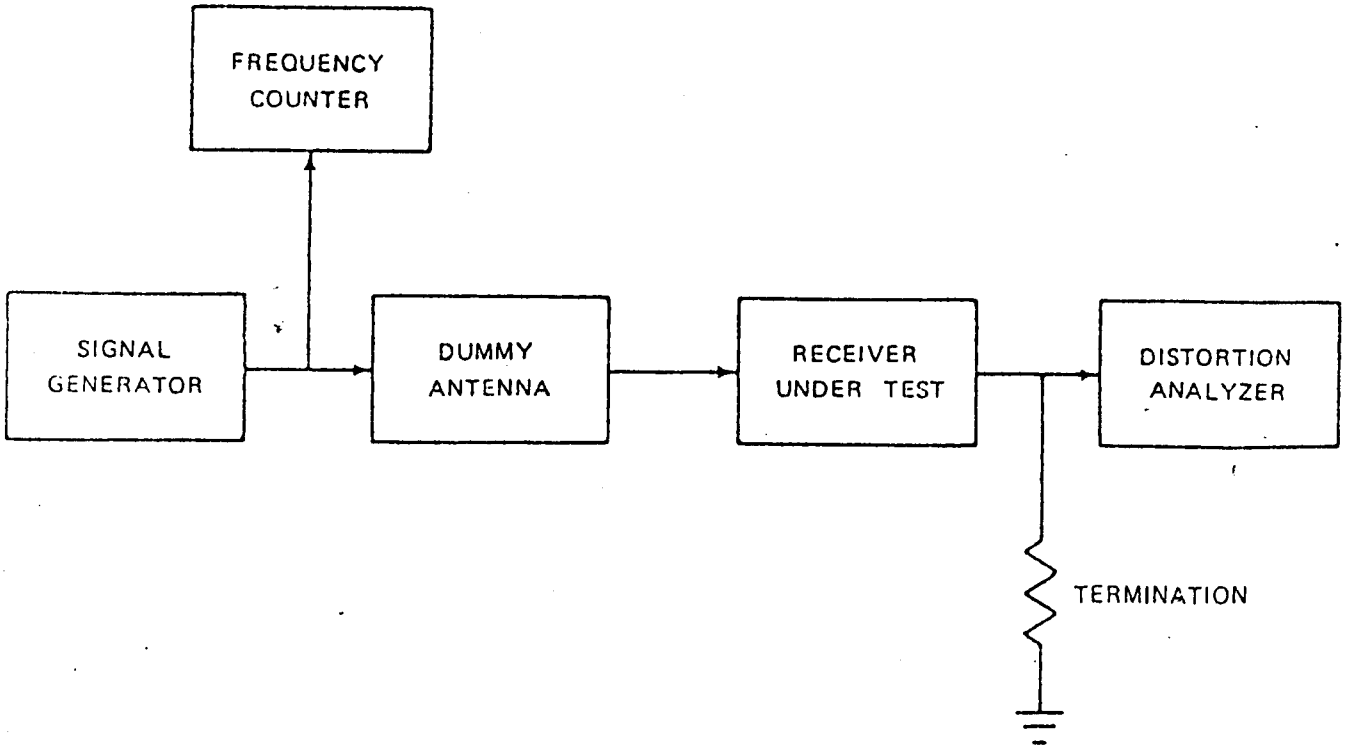


Figure 2. Sensitivity and selectivity measurement block diagram.

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NOTE: The selectivity as measured by this test gives an indication of overall gain and sensitivity at the center tuned frequency. The test shows that the response at frequencies slightly removed from the tuned frequency is a measure of the receiver's ability to discriminate against off-channel radiations and in reality is also a measure of the receiver's bandpass characteristics.

b. Perform a sensitivity test in accordance with paragraph 5.1.7.

c. Increase the signal generator power output by 3 decibels.

d. Tune the signal generator slowly below the receiver tuned frequency until the standard response, as defined in the selectivity measurement procedures of MIL-STD-449D, is re-established.

e. Note the signal generator input frequency and power output.

f. Tune the signal slowly above the tuned frequency until the standard response is again re-established.

g. Note the signal generator's frequency and power output.

h. Repeat the procedure for signal levels 6, 12, 20, 40, 60, and 80 decibels, above that established in step b, above.

5.1.9 Recording Equipment Test.

a. Apply a known signal to the input of the recorder with the recording equipment connected in its normal operational configuration.

b. Note and record the accuracy and sensitivity of the recorder.

c. Repeat step a above, applying at least 20 frequencies covering the range of the recorder.

d. Repeat step b, above.

NOTE: The recording equipment may consist of a frequency meter and a paper chart recorder which can be run at various paper speeds. Each type of signal (temperature, humidity, and pressure) has a frequency (and sequence) depending upon the value of the measurement.

5.1.10 Computer Test. Complete testing shall be conducted in accordance with appropriate sections of TOP 6-2-060, Data Processing Equipment, 15 December 1969, AD No. 866467,[9] and TOP 5-2-532, Computers (Electronic, 11 March 1968, AD No. 718236[10].

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NOTE: Some meteorological sounding systems use a computer to convert the received meteorological data to a more meaningful interpretation of such data. The primary advantage in this type of system is the speed with which the data can be converted to meaningful interpretations and the straight readout of the pertinent data.

Test computer portion of such a system by means of a prerecorded tape containing the necessary data to be processed. Record the information on the test tape and also record on the visual recorder in order to provide a basis for analysis of test results. The computer must select values which have a predetermined amount of change and other fixed values. These values can be manually measured on the visual chart and deviations between the input data and the output data determined and analysed accordingly.

5.1.11 Antenna Test. Antenna tests shall be performed in accordance with TOP 6-2-020, Radar Antenna Tests, 15 June 1981, AD No. A100393[11].

5.1.12 Balloon Test. Balloon tests shall be performed in accordance with TOP 6-2-182 Meterorological Equipment, Balloons, 1 March 1967, AD No. 718598[12] and TOP 6-2-184, Meteorological Equipment; Inflation, Tethering and Launching Equipment, 21 June 1968, AD No. 720580[13].

5.1.13 System Test.

a. Select one or two relatively well-known Radiosonde units from which the data received can be used as calibrated data.

b. Mount the known Radiosonde units and the Radiosonde test item on a supporting structure as shown in figure 3. Attach them to a vertical train hanging below the balloon and spaced an appropriate distance apart.

NOTE: Each Radiosonde unit shall transmit at a different frequency and the ground system shall be tuned to the specific frequency for the corresponding Radiosonde.

c. Set up two theodolites at a known distance apart, one being as close to the ground unit as possible.

d. Coordinate all operations involved, radiotheodolite-tracking, receiving-ground-unit, and balloon-release personnel, and ensure that all data is time referenced to release of the balloon.

e. Release the balloon assembly upon signal from the test director (see fig. 3) and commence tracking activities.

NOTE: The radiotheodolites shall resolve the distance, azimuth and elevation angles during this test.

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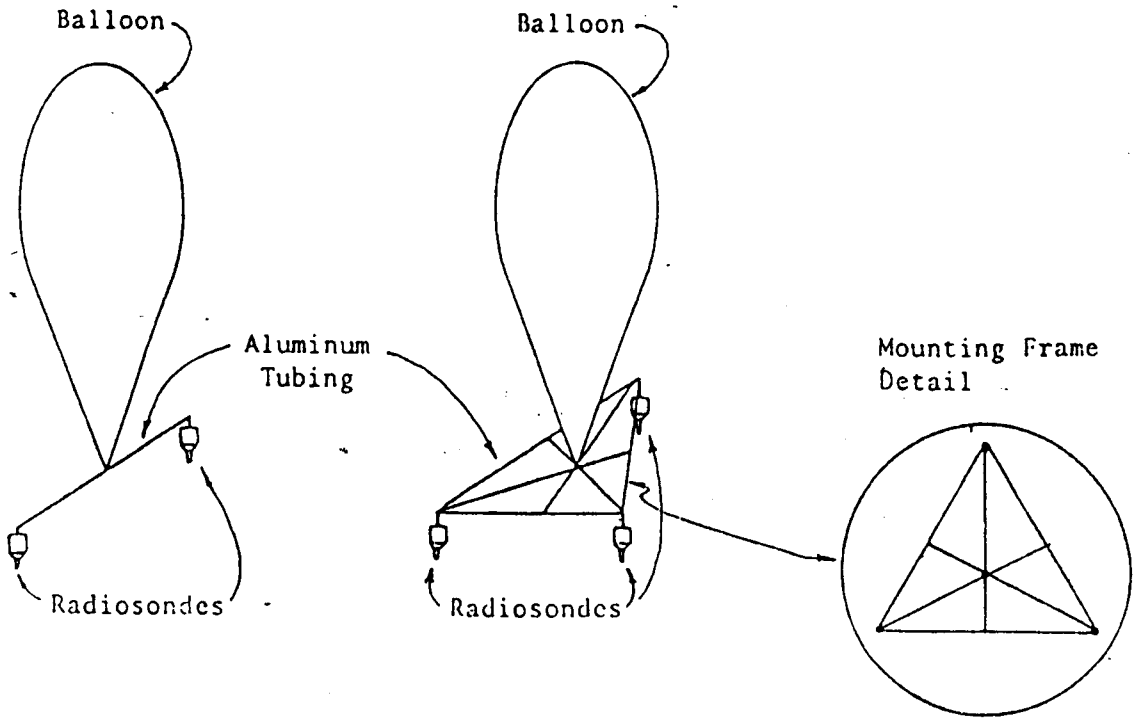


Figure 3. Mounting of radiosonde for systems test.

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5.2 Data Required.

5.2.1 Radiosonde Power Output Test.

a. Record the radiosonde's standard test frequency.

b. Record the signal generator output level and frequency.

c. Record the battery voltage at 5-minute intervals.

5.2.2 Radiosonde Battery Life Test.

a. Record battery voltage versus power output readings at 5-minute intervals.

b. Record temperature-altitude values.

5.2.3 Radiosonde Transmitter Range Test. Record data in accordance with TOP 6-2-242, Receiver-Transmitter, System Tests 15 June 1981, AD No. A100395.[6]

5.2.4 Sensor Response and Temperature, Humidity and Pressure Accuracy Test.

5.2.4.1 Temperature Sensor.

a. Reference sensor temperature and test item temperature indications at one-minute intervals.

b. Lag or response time of temperature element relative to reference sensor.

c. Radiosonde variations.

5.2.4.2 Humidity Sensor.

a. Reference sensor humidity indication versus test item humidity indication as a function of time.

b. Temperature.

c. Radiosonde variations.

5.2.4.3 Pressure System.

a. Aneroid barometer indication together with test item pressure indication over the range specified.

b. Radiosonde variations.

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5.2.5 Upper Atmosphere Wind Speed and Direction Measuring Accuracy Test.

Data shall be recorded as specified in TOP 6-2-070, in addition to the following:

a. The minimum distance for signal lock-on shall be recorded.

b. The minimum elevation angle for signal lock-on shall be recorded.

c. The azimuth and elevation angles and range for any meaningful signal effect shall be recorded.

d. The azimuth angle readings from the ground unit and the angles measured with the theodolite shall be recorded.

5.2.6 Frequency Accuracy and Stability Test. Record the data in accordance with TOP 6-2-517.

5.2.7 Receiver Sensitivity Test.

a. Test frequency for each step.

b. Generator output level in dbm.

5.2.8 Receiver Selectivity Test. Record signal generator's input frequency and power output at various signal levels.

5.2.9 Recording Equipment Test.

a. Frequency of the input signals.

b. Recorded output data.

5.2.10 Computer Test.

a. Data input.

b. Data output.

c. Other data as required by TOP's 6-2-060 and 5-2-532.

5.2.11 Antenna Test. Record the data as specified in TOP 6-2-020.

5.2.12 Balloon Test. Record the data in accordance with TOP 6-2-182 and TOP 6-2-184.

5.2.13 System Test.

a. Transmitting frequency of each Radiosonde.

b. Receiver data from each system.

c. Distance between theodolites.

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d. Azimuth and elevation angles from each theodolite.

6.0 DATA REDUCTION AND PRESENTATION.

Processing of raw test data shall, in general, consist of organizing, marking for correlation and identification, and grouping the test data according to subtest title. Test criteria or test item specifications shall be noted on the test data presentation to facilitate analysis and comparison. Where necessary, test data measurements shall be converted to be compatible with units given by test criteria or specifications.

6.1 Radisonde Power Output Test. Present the data in a suitable form for comparison with applicable criteria.

6.2 Radiosonde Battery Life Test. Present the data in a graphic form for comparison with applicable criteria.

6.3 Radiosonde Transmitter Range Test. Present the data for reduction and analysis.

6.4 Sensor Response and Temperature, Humidity and Pressure Accuracy Test.

Present the data in a suitable form for comparison with applicable criteria.

6.5 Upper Atmosphere Wind Speed and Direciton Measuring Accuracy Test.

a. Reduce and present the data as specified in TOP 6-2-070.

b. Present the data in a suitable form for comparison with applicable criteria.

6.6 Frequency Accuracy and Stability. Present the data in accordance with TOP 6-2-517.

6.7 Receiver Sensitivity Test. Present the data in a suitable form for comparison with applicable criteria.

6.8 Receiver Selectivity Test. Present the data in a suitable form for comparison with applicable criteria.

6.9 Recording Equipment Test. Present the sensitivity and accuracy data in a suitable form for comparison with applicable criteria.

6.10 Computer Test. Present the input and output data in a suitable form for comparison with applcable criteria.

6.11 Antenna Test.

Reduce and present the data as specified in TOP 6-2-020.

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6.12 Balloon Test. Reduce and present the data in accordance with TOP 6-2-182 and TOP 6-2-184.

6.13 System Test.

Reduce and present the data in a tabular form for comparison purposes.

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APPENDIX A
CHECKLISTS

A.1 PRETEST

a. Detailed test procedure available _____

b. Instrumentation support facility available _____

c. Instrument calibrated _____

d. Engineering logbook available _____

e. Logbook entries of date, test conductor _____

f. Safety precautions instituted _____

g. Test personnel informed of review rights _____

h. Test item sample size plan prepared _____

A.2 PERFORMANCE TESTS

a. Detailed test procedure available _____

b. Applicable technical manual(s) available _____

c. Test instrumentation calibrated _____

A.2.1 Radiosonde Power Output Test

a. Test item stabilized _____

b. Technical manual tuning procedure used _____

c. Signal substitution used _____

A.2.2 Radiosonde Battery Life Test

a. Temperature-altitude chamber checked _____

b. Voltage and power output readings taken at 5-minute intervals _____

c. Lowest voltage level for usable signal recorded _____

A.2.3 Radiosonde Transmitter Range Test

Procedures available _____

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A.2.4 Sensor Response and Temperature, Humidity, and Pressure Accuracy Test.

A.2.4.1 General

a. Reference sensors available _____

b. Chamber range verified _____

c. Radiosonde uniformity checked _____

d. Transmitter adjusted one element _____

A.2.4.2 Temperature Sensor

a. Temperature reference standard available _____

b. Readings made at one minute intervals _____

c. 5° C increments over entire range _____

d. Temperature element response time checked _____

A.2.4.3 Humidity Sensor

a. Reference standard available _____

b. 10% relative humidity increments to 100% _____

c. High temperature range checked _____

d. Low temperature range checked _____

e. Mid-temperature range checked _____

A.2.4.4 Pressure System

a. Element calibrated and adjusted _____

b. High range checked _____

c. Pressure decreased per calibration chart _____

d. Lowest range checked _____

A.2.5 Upper Wind Speed and Direction Measurement Accuracy Test

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a. TOP 6-2-070 available _____

b. Minimum lock-on range determined _____

c. Minimum elevation range determined _____

d. Critical reflection angles determined _____

e. Azimuth angle accuracy checked _____

A.2.6 Frequency Accuracy and Stability Test.

TOP 6-2-517 available _____

A.2.7 Receiver Sensitivity Test.

a. MIL-STD-449D available _____

b. Sheilded room available _____

c. Test setup properly grounded _____

d. Receiver and signal generator set _____

e. Signal generator modulation and output level set correctly _____

f. All test frequencies checked _____

A.2.8 Receiver Selectivity Test.

a. Sensitivity test performed satisfactorily _____

b. Standard response determined at frequencies corresponding to increased signal generator levels of 3dB. _____

c. Standard response determined at 6, 12, 20, 40, 60 and 80 decibels above original 3dB level. _____

A.2.9 Recording Equipment Test

a. Recorder accuracy and sensitivity checked _____

b. Minimum of twenty frequencies used _____

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A.2.10 Computer Test

TOP 6-2-060 and TOP 5-2-532 available _____

A.2.11 Antenna Test

TOP 6-2-020 available _____

A.2.12 Balloon Test

TOP 6-2-182 and TOP 6-2-184 available _____

A.2.13 System Test

a. Known units selected for calibrated data _____

b. Units mounted and frequencies set _____

c. Theodolites setup _____

d. All data time referenced to balloon release _____

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APPENDIX B
DATA COLLECTION SHEETS

B.1 Radiosonde Power Output Test.

Test Procedure: _____ Date: _____

Test Director: _____ Test Conductor: _____

Test Item Nomenclature: _____

Test Item SN: _____

Radiosonde Transmitter Standard Test Frequency _____

Radiosonde Carrier Frequency (NFIM) _____

NFIM	Indicator	Setting	Attenuation Factors

Signal Generator (Transmitter Carrier Frequency)

Indicator	Setting	Attenuation Factors

Comparison: (Narrative) _____

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B.2 Radiosonde Battery Life Test

Test Item Nomenclature: _____

Test Item SN: _____

	Battery #1		Battery #2		Battery #3	
Ambient	_____V	_____dBm	_____V	_____dBm	_____V	_____dBm
100,000 Ft. (Simulated)						
0 min.	_____V	_____dBm	_____V	_____dBm	_____V	_____dBm
5	_____V	_____dBm	_____V	_____dBm	_____V	_____dBm
10	_____V	_____dBm	_____V	_____dBm	_____V	_____dBm
15	_____V	_____dBm	_____V	_____dBm	_____V	_____dBm
20	_____V	_____dBm	_____V	_____dBm	_____V	_____dBm
25	_____V	_____dBm	_____V	_____dBm	_____V	_____dBm
30	_____V	_____dBm	_____V	_____dBm	_____V	_____dBm
35	_____V	_____dBm	_____V	_____dBm	_____V	_____dBm
40	_____V	_____dBm	_____V	_____dBm	_____V	_____dBm
45	_____V	_____dBm	_____V	_____dBm	_____V	_____dBm
50	_____V	_____dBm	_____V	_____dBm	_____V	_____dBm
55	_____V	_____dBm	_____V	_____dBm	_____V	_____dBm
60	_____V	_____dBm	_____V	_____dBm	_____V	_____dBm

Lowest Usable Voltage Level (Narrative) _____

_____

B.3 Radiosonde Transmitter Range Test

Test Item Nomenclature: _____

Test Item SN: _____

Include data collected per TOP 6-2-515, Data Collection Sheet(s).

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B.4 Sensor Response-Temperature, Humidity, Pressure Test

Test Item Nomenclature: _____

Test Item SN: (3) _____ / _____ / _____

Reference Sensor SN: _____

Reference Aneroid Barometer SN: _____

B.4.1 Temperature Sensor

Test Item SN: (3) _____ / _____ / _____

Temperature (5° C Increments)	Ref. Sensor	Test Sensor #1	Test Sensor #2	Test Sensor #3
Ambient ____°C				
1 min				
2 min				
3 min				
4 min				
5 min				
____°C				
1 min				
2 min				
3 min				
4 min				
5 min				
____°C				
1 min				
2 min				
3 min				
4 min				
5 min				

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Activate heating element - Record temp/response time:

	/	/	/	/
Continue				
Ambient ____°C				
1 min				
2 min				
3 min				
4 min				
5 min				
____°C				
1 min				
2 min				
3 min				
4 min				
5 min				
____°C				
1 min				
2 min				
3 min				
4 min				
5 min				

Activate heating element - Record temp/response time:

	/	/	/	/
Continue				

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Activate heating element - Record temp/response time:

Continue	/	/	/	/
Ambient ____°C				
1 min				
2 min				
3 min				
4 min				
5 min				
____°C				
1 min				
2 min				
3 min				
4 min				
5 min				
____°C				
1 min				
2 min				
3 min				
4 min				
5 min				

Activate heating element - Record temp/response time:

	/	/	/	/
Continue				

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Activate heating element - Record temp/response time:

	/	/	/	/
Continue				
Ambient ____°C				
1 min				
2 min				
3 min				
4 min				
5 min				
____°C				
1 min				
2 min				
3 min				
4 min				
5 min				
____°C				
1 min				
2 min				
3 min				
4 min				
5 min				

Activate heating element - Record temp/response time:

	/	/	/	/
Continue				

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Activate heating element - Record temp/response time:

	/	/	/	/
Continue				
Ambient ____°C				
1 min				
2 min				
3 min				
4 min				
5 min				
____°C				
1 min				
2 min				
3 min				
4 min				
5 min				
____°C				
1 min				
2 min				
3 min				
4 min				
5 min				

Activate heating element - Record temp/response time:

	/	/	/	/
Continue				

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Ambient _____°C				
1 min				
2 min				
3 min				
4 min				
5 min				

Test Narrative _____

_____

B.4.2 Humidity Sensor

Test Item SN: (3) _____ / _____ / _____

% Relative Humidity (Approximate)	High Temperature Range			
	Ref. Sensor	Test Sensor #1	Test Sensor #2	Test Sensor #3
0%				
10%				
20%				
30%				
40%				
50%				
60%				
70%				
80%				
90%				
100%				

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Low Temperature Range

%Relative Humidity (Approximate)	Ref. Sensor	Test Sensor #1	Test Sensor #2	Test Sensor #3
0%				
10%				
20%				
30%				
40%				
50%				
60%				
70%				
80%				
90%				
100%				

Mid Temperature Range

% Relative Humidity (Approximate)	Ref. Sensor	Test Sensor #1	Test Sensor #2	Test Sensor #3
0%				
10%				
20%				
30%				
40%				
50%				
60%				
70%				
80%				
90%				
100%				

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Test Narrative _____

_____

B.4.3 Pressure System

Test Item SN: (3) _____/_____

	Aneroid Barometer	Test Sensor #1	Test Sensor #2	Test Sensor #3
Highest Range				
Increment 1				
Increment 2				
Increment 3				
Increment 4				
Increment 5				
Increment 6				
Increment 7				
Increment 8				
Increment 9				
Increment 10				
Lowest Range				

Test Narrative _____

_____

B.5 Wind Speed/Direction Measurement Accuracy Test

Test Item SN: _____

Minimum Lock-on Range (<100 ft) (<30.5 m) _____

Minimum Lock-on Elevation _____

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	Range 1 (min)	Azimuth Angle	Range 2 (mid)	Azimuth Angle	Range 3 (max)	Azimuth Angle
0°						
15°						
30°						
45°						
60°						
75°						
90°						
75°/105°						
60°/120°						
45°/135°						
30°/150°						
15°/165°						
0°/180°						

Low Angle Effect (Narrative) _____

_____

B.6 Frequency Accuracy and Stability Test

Test Item SN: _____

Include data collected per TOP 6-2-517, Data Collection Sheet(s).

B.7 Receiver Sensitivity Test

Test Item SN: _____

Receiver Load Impedance: _____

System Modulation: _____

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	Signal Generator Output Level	Signal Generator Output Frequency
Sound Element Frequency		
Low Receiver Frequency		
Mid Receiver Frequency		
High Receiver Frequency		

Test Narrative _____

_____

B.8 Receiver Selectivity Test

Test Item SN: _____

Receiver Tuned Frequency: _____

(Test B.7 +:)	Signal Generator Input Frequency	Signal Generator Power Output
+3 dB		
+6 dB		
+12 dB		
+20 dB		
+40 dB		
+60 dB		
+80 dB		

Test Narrative _____

_____

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B.9 Recording Equipment Test

Test Item SN: _____

Recorder Frequency	Accuracy	Sensitivity
Frequency #1		
Frequency #2		
Frequency #3		
Frequency #4		
Frequency #5		
Frequency #6		
Frequency #7		
Frequency #8		
Frequency #9		
Frequency #10		
Frequency #11		
Frequency #12		
Frequency #13		
Frequency #14		
Frequency #15		
Frequency #16		
Frequency #17		
Frequency #18		
Frequency #19		
Frequency #20		

Test Narrative _____

_____

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B.10 Computer Test

Test Item SN: _____

Include data collected per TOP 6-2-060 and TOP 5-2-532, Data Collection Sheets(s).

B.11 Antenna Test

Test Item SN: _____

Include data collected per TOP 6-2-020, Data Collection Sheet(s).

B.12 Balloon Test

Test Item SN: _____

Include data collected per TOP 6-2-182 and TOP 6-2-184, Data Collection Sheet(s).

B.13 System Test

Test Item SN: _____

Reference Item SN: (2) _____ / _____

Theodolite Distance _____

Balloon Release Time _____

Specific Data	Ref Unit #1	Ref Unit #2	Test Unit
Temperature			
Humidity			
B. Pressure			
Wind Speed			
Wind Direction			

Test Narrative: _____

_____

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APPENDIX C
REFERENCES

1. TECOM Pam 310-4 — Index of Test Operations Procedures
2. TECOM Supplement 1 w/change 1, to AMC Reg 310-6 — Quality Assurance Publiction
3. TECOM Pam 70-3 — Project Engineers' Handbook
4. TECOM Reg 70-24 — Documenting TECOM Testing
5. TOP 3-1-002 AD No. 718229 25 Jan 67 — Confidence Intervals and Sample Size.
6. TOP 6-2-242 AD No. A100395 15 June 81 — Receiver-Transmitter System Tests
7. TOP 6-2-517 AD No. A100394 31 Dec 80 — Transmitter Frequency Accuracy and Stability
8. MIL-STD-449D 22 Feb 73 — Radio Frequency Spectrum Characteristics, Measurements of
9. TOP 6-2-060 AD No. 866467 15 Dec 69 — Data Processing Equipment
10. TOP 5-2-532 AD No. 718236 11 Mar 68 — Computers (Electronic)
11. TOP 6-2-020 AD No. A100393 15 June 81 — Radar Antenna Tests
12. TOP 6-2-182 AD No. 718598 1 Mar 67 — Meteorological Equipment, Balloons
13. TOP 6-2-184 AD No. 720580 21 June 68 — Meteorological Equipment, Inflation Tethering and Launching Equipment

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APPENDIX D
ABBREVIATIONS

CTP	Coordinated Test Program
LR	Letter Requirement
NFIM	Noise and Field Intensity Meter
ROC	Required Operational Capability

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